



CDS6224-STATISTICAL DATA ANALYSIS

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Introduction

DATASET :

<https://www.kaggle.com/datasets/ayeshasiddiq123/student-health>

WHY THIS?

This dataset analyzes student mental health and burnout. It offers a thorough understanding of the relationships between academic achievement and mental health outcomes. This report aims to discover a trend in sleeping hours across students from various academic years, and how it relates to academic performance.

NUMERICAL VARIABLES :

Sleeping_hours
academic_performance

CATEGORICAL VARIABLE

Academic_year

Part A: Exploratory Descriptive Analysis

Code

```
#mean
sleep_mean<-mean(student_sleep)
perform_mean<-mean(student_perform)

#median
sleep_median<-median(student_sleep)
perform_median<-median(student_perform)

#mode
get_mode <- function(v) {
  uniqv <- unique(v)
  uniqv[which.max(tabulate(match(v, uniqv)))]
}
sleep_mode<-get_mode(student_sleep)
perform_mode<-get_mode(student_perform)

#standard deviation
sleep_sd<-sd(student_sleep)
perform_sd<-sd(student_perform)

#variance
sleep_var<-var(student_sleep)
perform_var<-var(student_perform)
```

```

#range
range(student_sleep, na.rm = FALSE, finite = FALSE)
range(student_perform, na.rm = FALSE, finite = FALSE)

#quartiles
sleep_quart<-quantile(student_sleep)
perform_quart<-quantile(student_perform)

#IQR
sleep_iqr<-IQR(student_sleep)
perform_iqr<- IQR(student_perform)

#CV
sleep_cv<-(sleep_sd/sleep_mean)*100
perform_cv<-(perform_sd/perform_mean)*100

#graphs
sleep_boxplot<-boxplot(student_data$sleep_hours, main="Boxplot of Sleeping
Hours",col="pink")
perform_boxplot<-boxplot(student_data$academic_performance, main="Boxplot of
Academic Performance", col="lightblue")

sleep_hist<-hist(student_data$sleep_hours, main="Histogram Of Sleeping Hours",
xlab="Hours Of Sleep", col="pink")
perform_hist<-hist(student_data$academic_performance, main="Histogram Of
Academic Performance", xlab="Academic Score", col="lightblue")

```

Summary Statistics

	Sleeping_hours		academic_performance
Mean	6.4896		70.9484
Median	6.4778		71.0013
Mode	3		68.4111
Standard Deviation	1.47455		5.631026
Variance	2.174297		31.70845
Range	3 10		52.43488 88.79248
Quartiles	0%	3.000000	52.43488
	25% Q1	5.465457	67.11888
	50% Q2	6.477830	71.00127

	75% Q3	7.516024	74.75911
	100%	10.000000	88.79248
IQR	2.050567		7.640227
CV	22.72174		7.936787

Data Consistency

	Sleeping_hours	academic_performance
Standard Deviation	1.47455	5.631026
CV	22.65%	7.97%

Analysis :

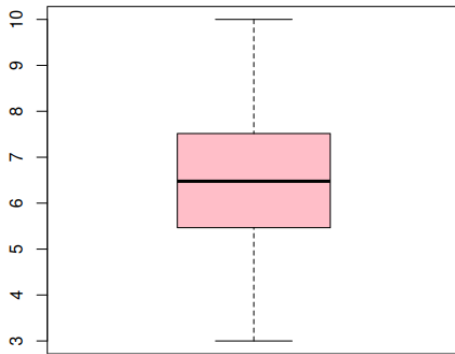
Sleeping Hours shows lower standard deviation ($s=1.47455$), meaning students' sleep duration is relatively closer to its mean ($\bar{x}=6.489601$) whereas academic performance with higher SD ($s=5.631026$) making it slightly further from its mean ($\bar{x}=70.94844$).

However, academic performance achieves lower cumulative variability ($CV=7.97\%$) resulting in more consistency and less variability to its mean compared to sleeping hours ($CV=22.65\%$).

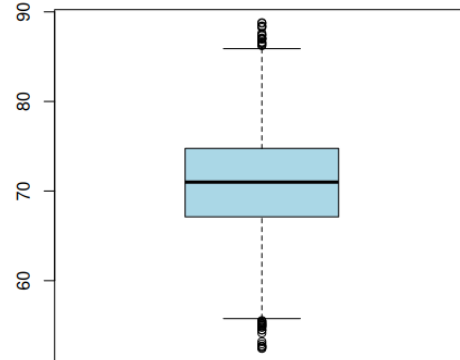
In conclusion, academic performance is much more consistent compared to sleeping hours.

Supporting Graphs

Boxplot of Sleeping Hours



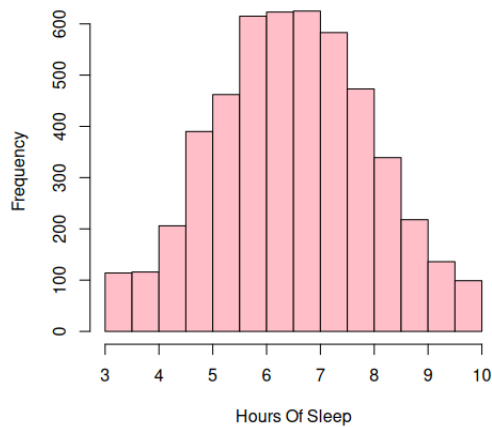
Boxplot of Academic Performance



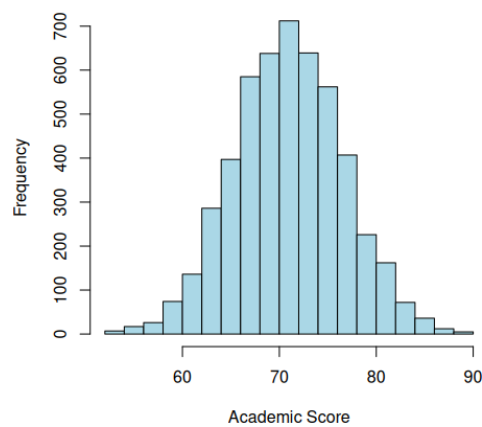
Analysis:

Although the boxplot of sleeping hours has a narrower absolute spread (lower SD), the CV reveals that academic performance is relatively more consistent to its mean. This is supported by its CV(7.97%) being much lower than that of Sleeping Hours (22.65%).

Histogram Of Sleeping Hours



Histogram Of Academic Performance



Analysis

The histogram of sleeping hours appears approximately bell-shaped and symmetry suggesting conformity to normal distribution with no visible outliers nor skewness. Sleep duration is scattered relative to its mean.

The histogram of academic performance follows an approximate normal bell shape and is symmetrical, its distribution is proportionally tighter around the mean despite outliers. Performance is more consistent relative to its mean.

So, even though the boxplot showed major outliers in performance, the histogram showed that most students cluster tightly around the mean therefore supporting the claim of greater consistency.

Part B: Distributional Assessment

Code

```
mu<-mean(student_perform)
sigma<- sd(student_perform)
#empirical rule
lower_68<-mu-sigma
upper_68<-mu+sigma
prop_68<-mean(student_perform>=lower_68 &
student_perform<=upper_68, na.rm=TRUE)
cat("68% of students scores between", lower_68 , " and", upper_68, "
\n")

lower_95<-mu-2*sigma
upper_95<-mu+2*sigma
prop_95<-mean(student_perform>=lower_95 &
student_perform<=upper_95, na.rm=TRUE)
cat("95% of students scores between", lower_95 , " and", upper_95, "
\n")

lower_997<-mu-3*sigma
upper_997<-mu+3*sigma
prop_997<-mean(student_perform>=lower_997 &
student_perform<=upper_997, na.rm=TRUE)
cat("99.7% of students scores between", lower_997 , " and", upper_997, "
\n")

#qqplot
```

```
perform_qqplot<-qqnorm(student_data$academic_performance, main="QQPlot
Of Academic Performance", col="lightblue")
qqline(student_data$academic_performance, col = "darkblue", lwd = 2)

#shapiro
shapiro.test(sample(student_data$academic_performance))
```

Empirical Formula

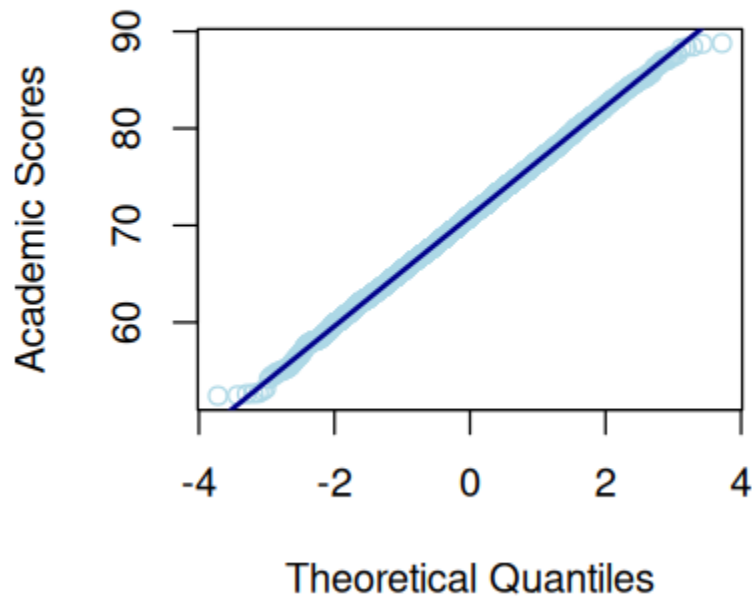
	Lower	Upper	Proportion
68%	65.31741	76.57946	0.6827365 (68.27%)
95%	59.68638	82.21049	0.9545909 (95.45%)
99.7%	54.05536	87.84151	0.9975995 (99.75%)

Analysis :

Based on the empirical rule, the variable academic performance seems to be normally distributed. 68.27% of the observations fall within one standard deviation, 95.45% within two, and 99.75% within 3. These results closely follow the 68-95-99.7 rule, which indicates normality.

QQ-Plot

QQPlot Of Academic Performance



Analysis

Based on the QQPlot, the variable academic performance appears to be normally distributed as data points are clustered around the reference line with little deviations at the tails. This indicates the distribution of academic scores is approximately normal. This proves normality.

Shapiro-Wilk Normality Test

W	0.99965
P-Value	0.5574

Analysis

The Shapiro-Wilk test resulted in a p-value greater than 0.005 (p-value = 0.2954) and $W=0.99965$. Since the P-value is greater than 0.05, it confirms that the data is normally distributed.

Method Analysis

Among the methods used, the Shapiro-Wilk test is the most reliable as it provides a statistical test of normality, which results in $W=0.99965$ and $p=0.5574$ in this case, which furthermore supports the claim of normality. However, QQ plots are useful to visually support the bell-shaped distribution and alignment with theoretical quantiles. Therefore, applying the Shapiro-Wilk test and QQ plots are reliable and provide strong evidence.

Part C: Bivariate Relationship Analysis

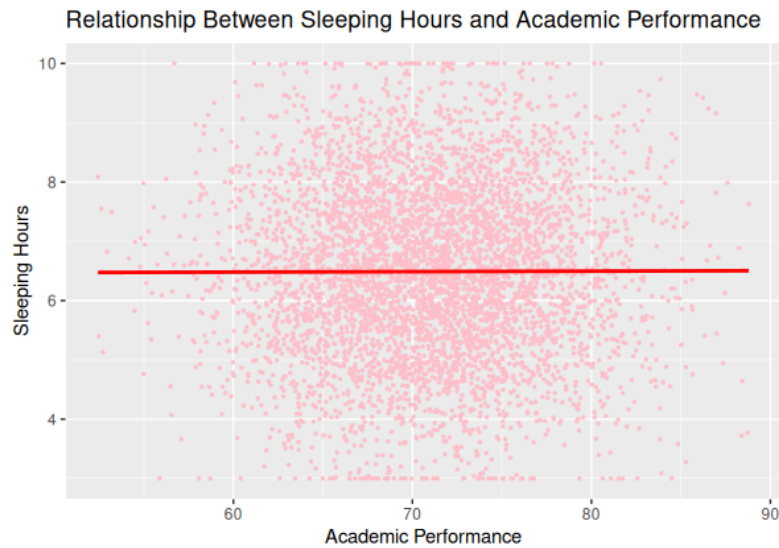
Code

```
df<-data.frame(x=student_perform, y=student_sleep)
ggplot(df,
aes(x=x,y=y))+geom_point(col="pink",size=0.5)+geom_smooth(method="lm",se=FALSE, col="red")+labs(title="Relationship Between Sleeping Hours and Academic Performance", x="Academic Performance",y="Sleeping Hours")

#pearson
#1 calculate sums of square
Sxx<-sum((student_perform-mean(student_perform))^2)
Syy<-sum((student_sleep-mean(student_sleep))^2)
Sxy<-sum((student_perform-mean(student_perform))*(student_sleep-mean(student_sleep)))
cat("")
#2 calculate Pearson correlation coefficient (r)
r<-Sxy/sqrt(Sxx*Syy)

#3 coefficient of determination
r_squared<-format(r^2,scientific=FALSE)
```

Scatter Plot



Pearson Correlation Coefficient

Variable	Value
S_{xx}	158478.8
S_{yy}	10867.14
S_{xy}	125.3834
r^2	0.000009128374
r	0.00302132

Analysis :

From the look of the scatter plot, sleeping hours and academic performance are largely independent from one another in this dataset. The trend line lays flat indicating no significant relationship.

The Pearson correlation coefficient resulted in $r=0.003$ ($r^2 \approx 0.000009$) which is extremely close to 0, indicating no relationship which supports the claim that both variables are independent and having weak relationships. This is also supported by the result of covariance ($S_{xy}=125.38$) which is positive but miniscule in relative to its S_{xx} and S_{yy} score. Therefore, the relationship between the variables sleeping hours and academic performance is neither causal nor associative.

Part D: Application & Critical Reflection

Insights

According to this dataset, we observe the relationship between sleeping hours and academic performance. Numbers show that the relationship is almost 0, which means that how much a student sleeps doesn't predict how well they perform academically. This claim is supported by the correlation value ($r=0.003$) being so close to 0 that it has no measurable effect on scores whether you sleep more or less. This suggests that other factors like study hours, motivation have more significant ties with academic performance.

How Inferential Statistics Help

Descriptive statistics like (mean, median, mode, etc) summarizes the data that's available. It tells you what that dataset looks like, but does not prove that the results could be applied beyond that. Inferential statistics are helpful to make conclusions beyond the sample - it's used to make predictions, inference on the larger population. It utilizes methods like hypothesis tests, confidence intervals, and regression to assess whether the patterns in a sample could be applied to the population.

Power Of Visualizations

Data visualizations turn numbers into visual insights which eases comprehension, pattern recognition and understanding. It emphasizes trends, anomalies, errors and relationships that may not be noticeable in a spreadsheet of just numbers. Visuals also help with providing more details on where, and when it happened, supporting better communication and engagement for the audience.